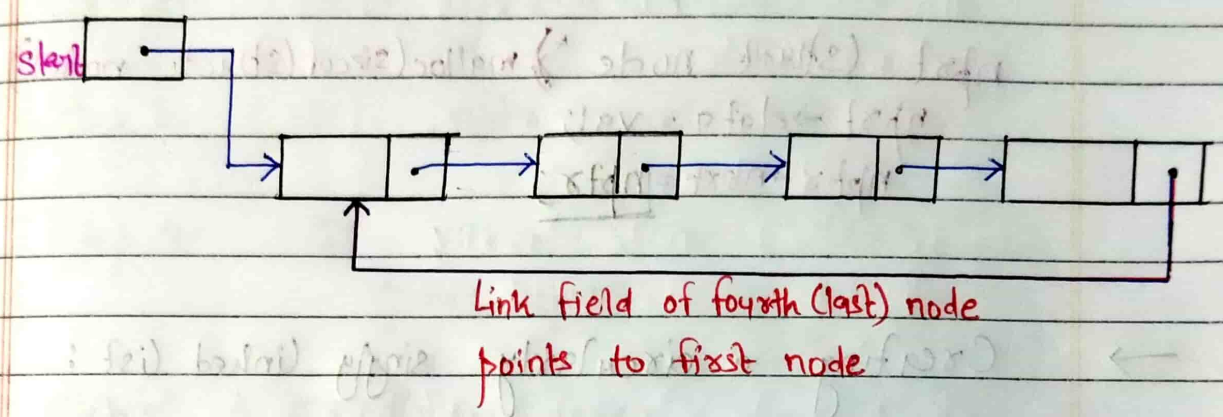
Circularly Singly Linked List

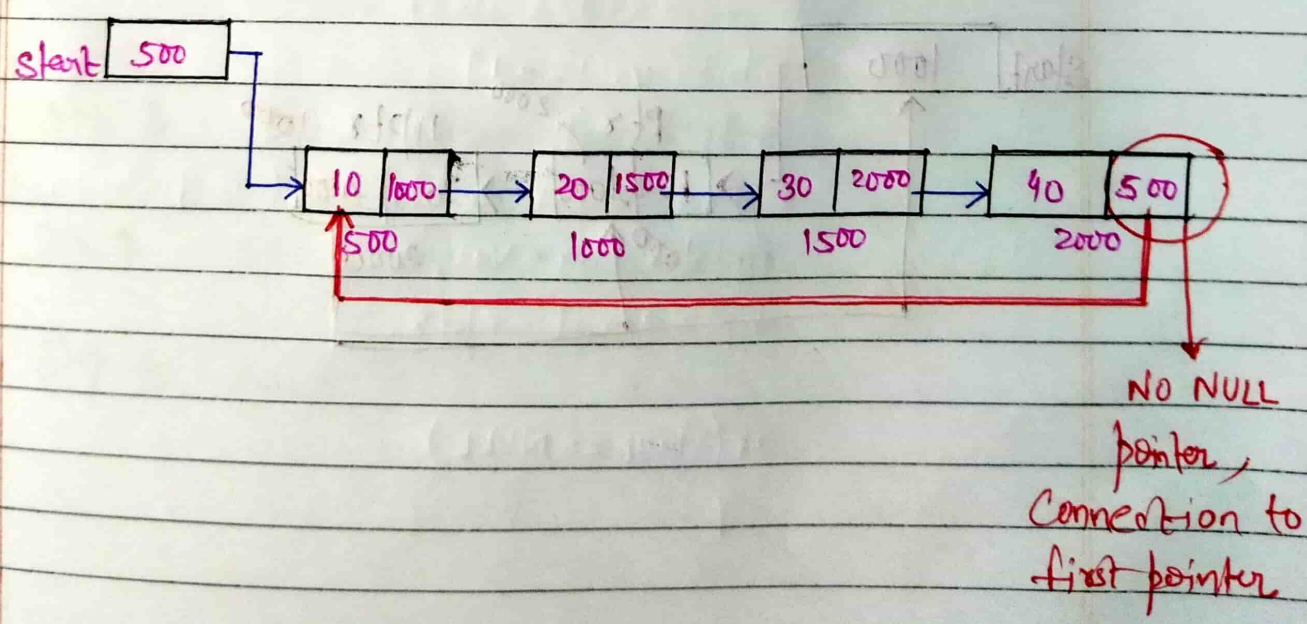
A circularly singly linked list is a variation of a singly linked list where the last node points back to the the first node instead of having a NULL pointer.



→ structure of node
→ creating of new node [little change]
→ creating of head pointer (start)
→ creating new node dynamically

} Same as singly linked list

→ Eg;



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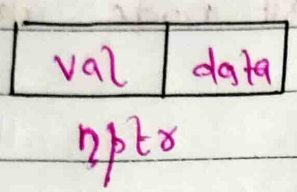
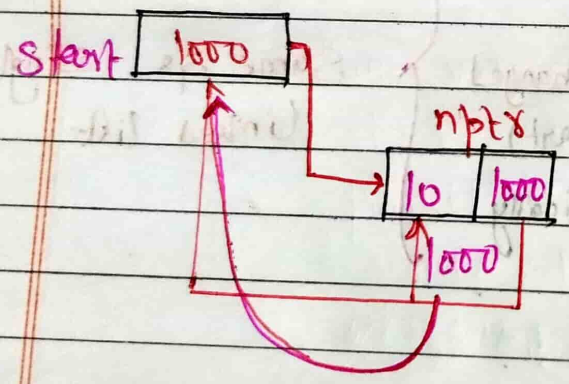
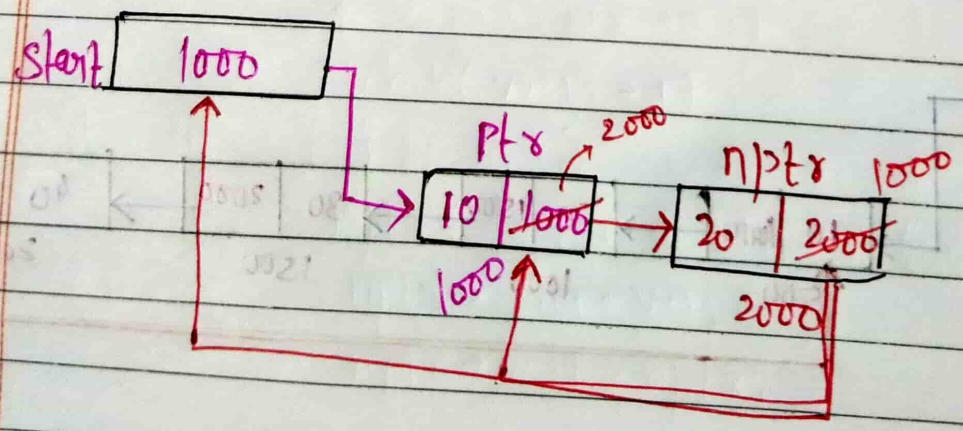
→ Syntax of creating a new node:




```
nptr = (struct node *) malloc(sizeof(struct node));
nptr->data = val;
nptr->next = nptr;
```

→ Creating a circularly singly linked list:





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→ Algorithm for creating a circularly singly linked list :-

Step 1. Create nptr [new_node]

Step 2. set nptr → data = val

Step 3. set nptr → next = nptr

Step 4. IF VAL = -1
goto step 10

Step 5. IF (start == NULL)
5.1) start = nptr;
nptr = start;

Step 6. Else,
6.1. set ptr = start

Step 7. Repeat until ptr → next != start
7.1. ptr = ptr → next

Step 8. set ptr → next = nptr

Step 9. set nptr → next = start

Step 10. Exit

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// C program to create a circularly singly
// linked list

```
#include<stdio.h>
#include<stdlib.h>
struct node
{
    int data;
    struct node *next;
};

struct node *create_list(struct node *start)
{
    int val;
    struct node *ptr, *nptr;
    printf("enter -1 to end\n");
    scanf("%d", &val);

    while(val!=-1)
    {
    nptr=(struct node*)malloc(sizeof(struct node));
    nptr->data=val;
    nptr->next=nptr;

    if(start==NULL)
    {
```

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```
    start = nptr;
    nptr = start;          // for creating circularly
  }
  else
    {
        ptr = start;
        while(ptr->next != start)
          {
              ptr = ptr->next;
          }
          ptr->next = nptr;
          nptr->next = start;  // for creating circularly
      }
        printf("enter data");
        scanf("%d", &val);
    }
    return start;
}

int main()
  {
      struct node *start = NULL;
      start = create_list(start);
      return 0;
  }
```

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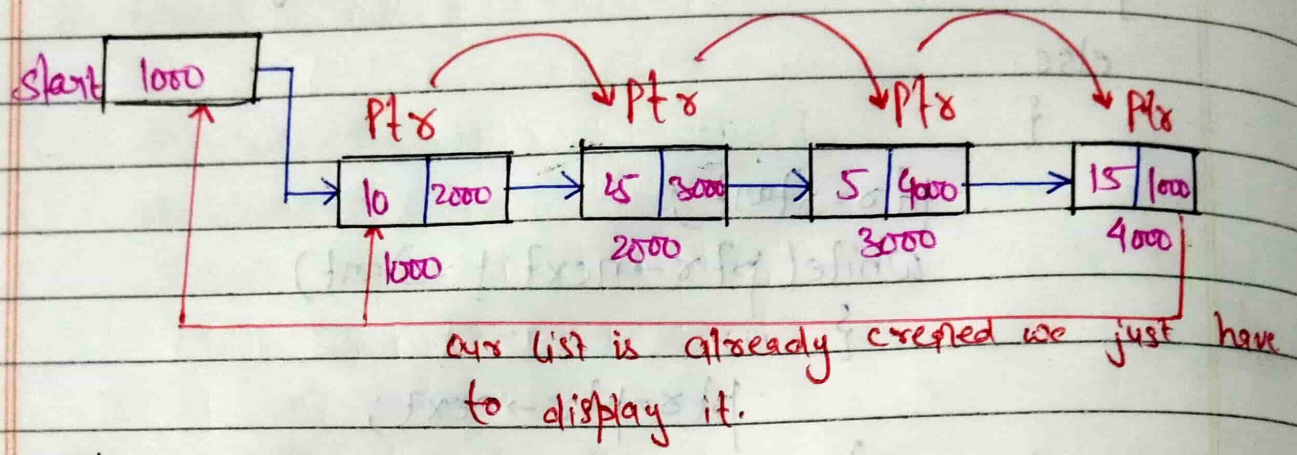
→ Algorithm for display a cricularly singly linked list:



Step 1.

IF Start == NULL

print "empty list"

goto step 5.

step 2. Else,

Set ptr = Start

step 3. do,

print ptr → data

3.1. ptr = ptr → next

Step 4. Repeat until ptr != Start

Step 5. Exit

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```
// C program for creating and display
// circular singly linked list

#include<stdio.h>
#include<stdlib.h>
struct node
  {
      int data;
      struct node *next;
  };
    struct node *create_list(struct node *start)
    {
        int val;
        struct node *ptr, *nptr;
        printf("enter -1 to end\n");
        scanf("%d", &val);
        while( val != -1 )
          {
              nptr=(struct node *)malloc(sizeof(
                struct node));
              nptr->data=val;
              nptr->next=nptr;

              if(start==NULL)
                {
                    start = nptr;
                    nptr = start;
                }
              else
```

```
{
    ptr = start
    while(ptr -> next != start)
        {
            ptr = ptr -> next;
        }
        ptr -> next = nptr;
        nptr -> next = start;
    }
    printf("enter data");
    scanf("%d", &val);
}
return start;
}

struct node *display_list(struct node *start)
{
    struct node *ptr;
    if(start == NULL)
        {
            printf("empty list");
        }
    else
        {
            ptr = start;
            do
                {
                    printf("%d", ptr -> data);
                    ptr = ptr -> next;
                }
```

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```
    while( ptr1 = start );
  }
    return start;
}

int main ()
  {
      struct node *start = NULL;
      start = create_list (start);
      start = display_list(start);
      return 0;
  }
```

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